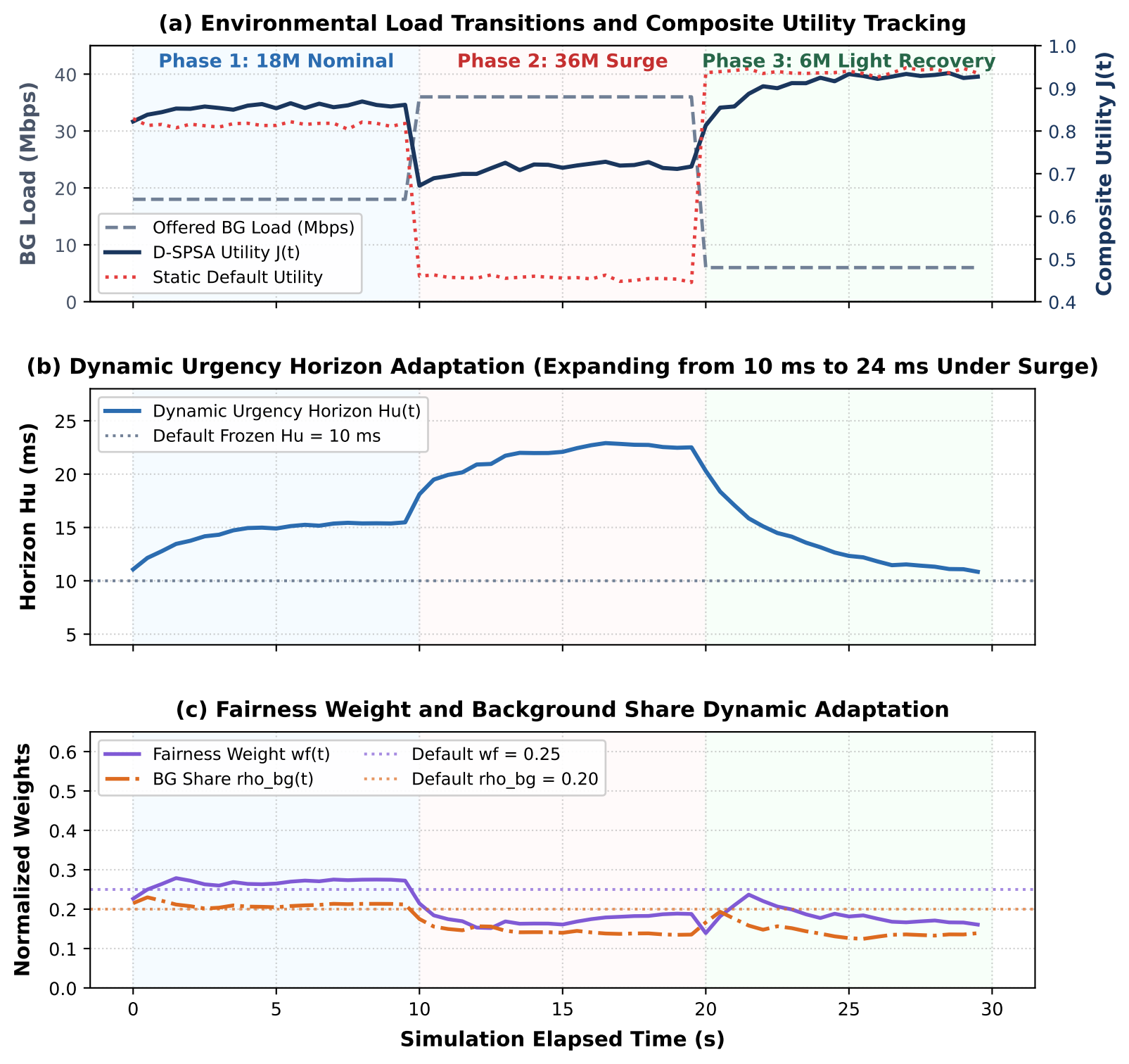
(a) Environmental Load Transitions and Composite Utility Tracking
Phase 1: 18M Nominal
Phase 2: 36M Surge
Phase 3: 6M Light Recovery
BG Load (Mbps)
Composite Utility J(t)
Offered BG Load (Mbps)
D-SPSA Utility J(t)
Static Default Utility
(b) Dynamic Urgency Horizon Adaptation (Expanding from 10 ms to 24 ms Under Surge)
Horizon Hu (ms)
Dynamic Urgency Horizon Hu(t)
Default Frozen Hu = 10 ms
(c) Fairness Weight and Background Share Dynamic Adaptation
Normalized Weights
Fairness Weight wf(t)
BG Share rho_bg(t)
Default wf = 0.25
Default rho_bg = 0.20
Simulation Elapsed Time (s)